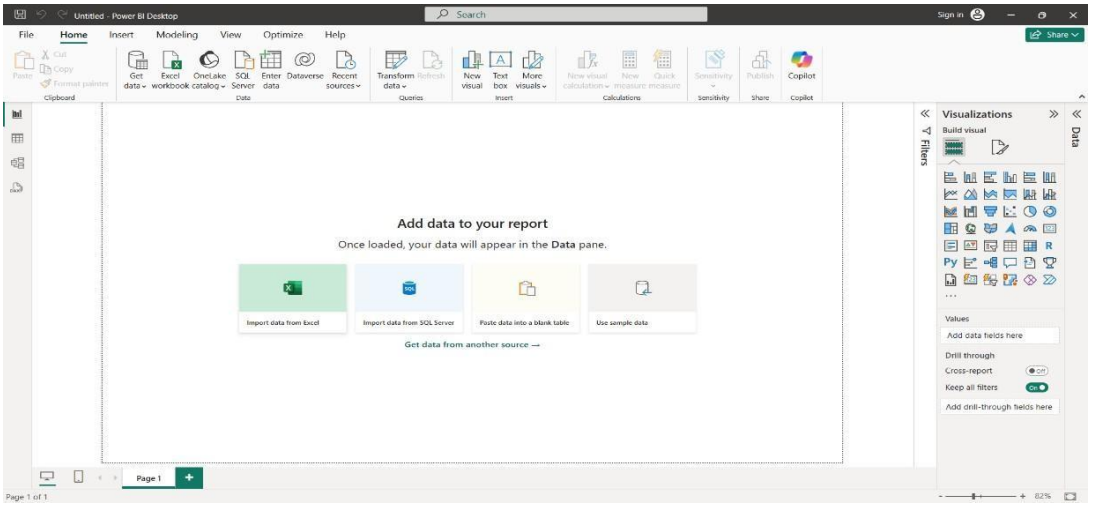
EX.NO: 7	IMPLEMENTATION OF DECISION TREE USING R TOOL
DATE:	

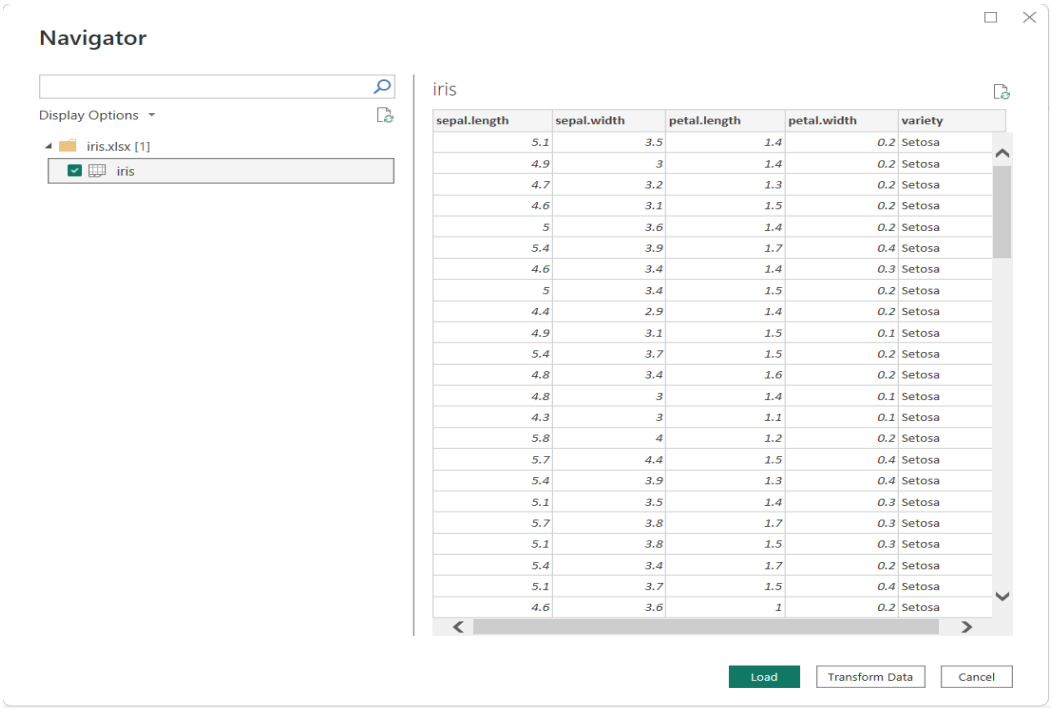
AIM:

To implement the Decision Tree using R programming in Power BI.

PROCEDURE:

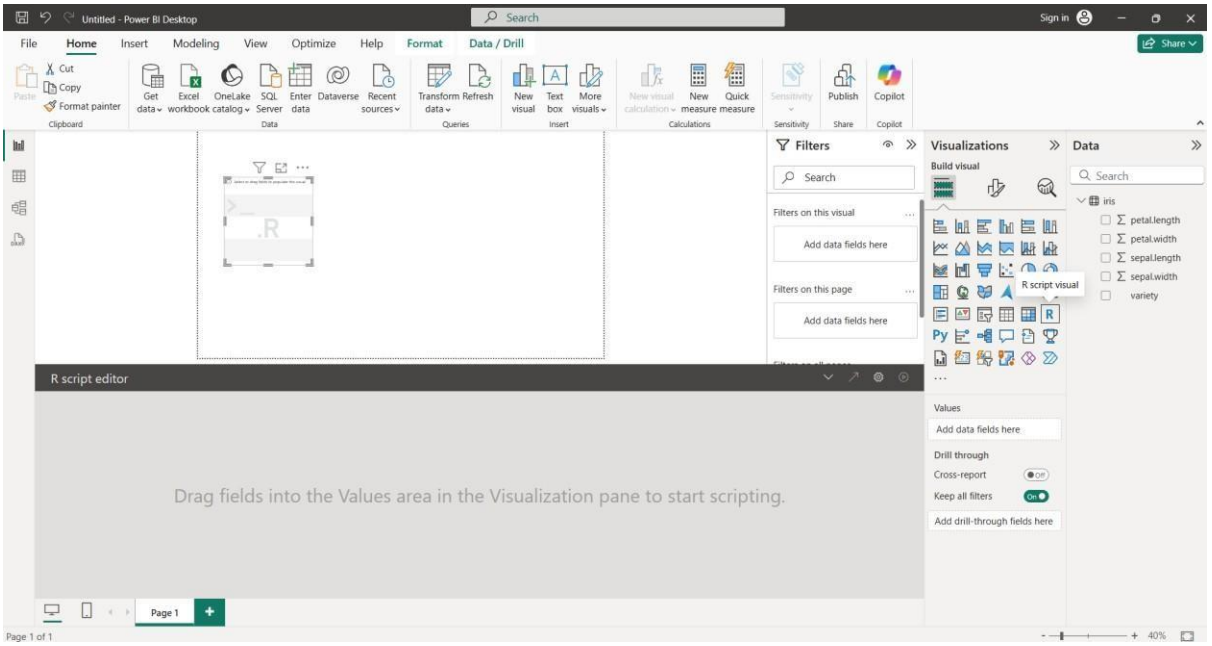
Step 1: Import The Dataset into Power BI



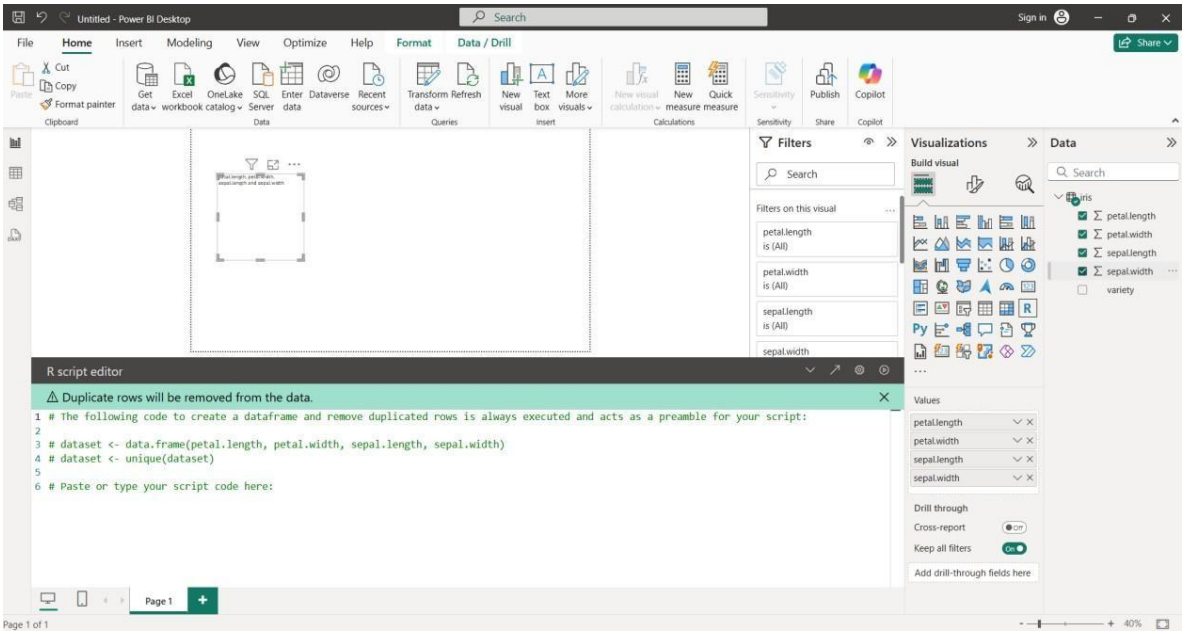


sepal.length	sepal.width	petal.length	petal.width	variety
5.1	3.5	1.4	0.2	Setosa
4.9	3	1.4	0.2	Setosa
4.7	3.2	1.3	0.2	Setosa
4.6	3.1	1.5	0.2	Setosa
5	3.6	1.4	0.2	Setosa
5.4	3.9	1.7	0.4	Setosa
4.6	3.4	1.4	0.3	Setosa
5	3.4	1.5	0.2	Setosa
4.4	2.9	1.4	0.2	Setosa
4.9	3.1	1.5	0.1	Setosa
5.4	3.7	1.5	0.2	Setosa
4.8	3.4	1.6	0.2	Setosa
4.8	3	1.4	0.1	Setosa
4.3	3	1.1	0.1	Setosa
5.8	4	1.2	0.2	Setosa
5.7	4.4	1.5	0.4	Setosa
5.4	3.9	1.3	0.4	Setosa
5.1	3.5	1.4	0.3	Setosa
5.7	3.8	1.7	0.3	Setosa
5.1	3.8	1.5	0.3	Setosa
5.4	3.4	1.7	0.2	Setosa
5.1	3.7	1.5	0.4	Setosa
4.6	3.6	1	0.2	Setosa

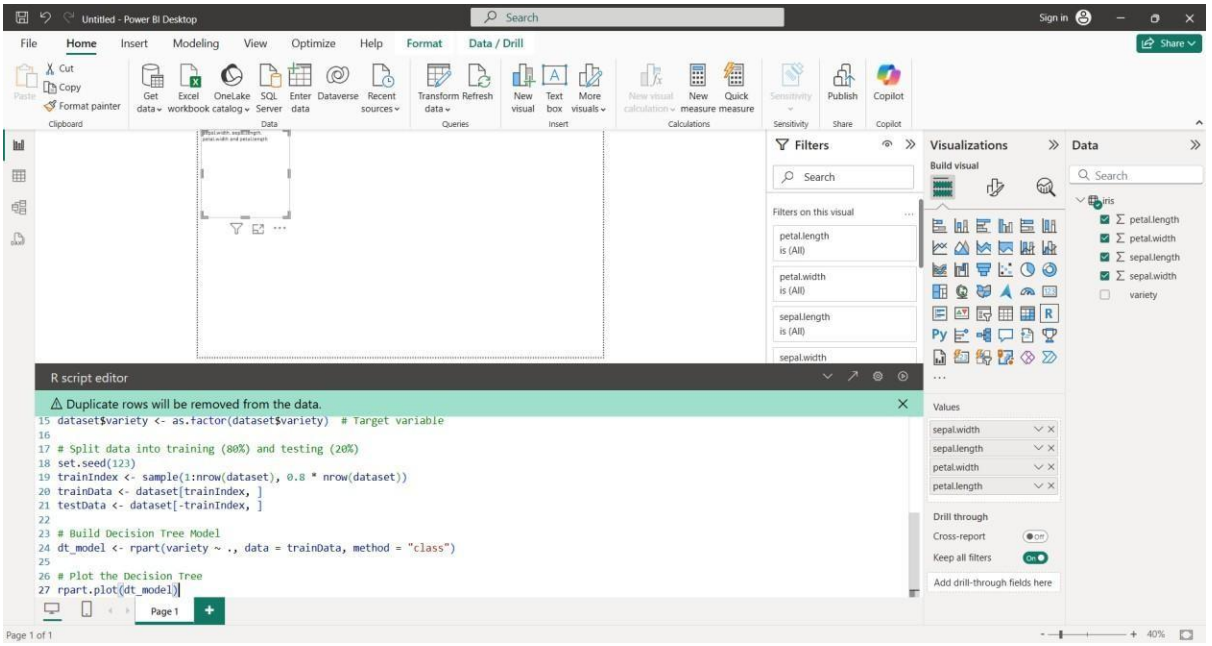
Step 2: Then select R script visual from the Visualization



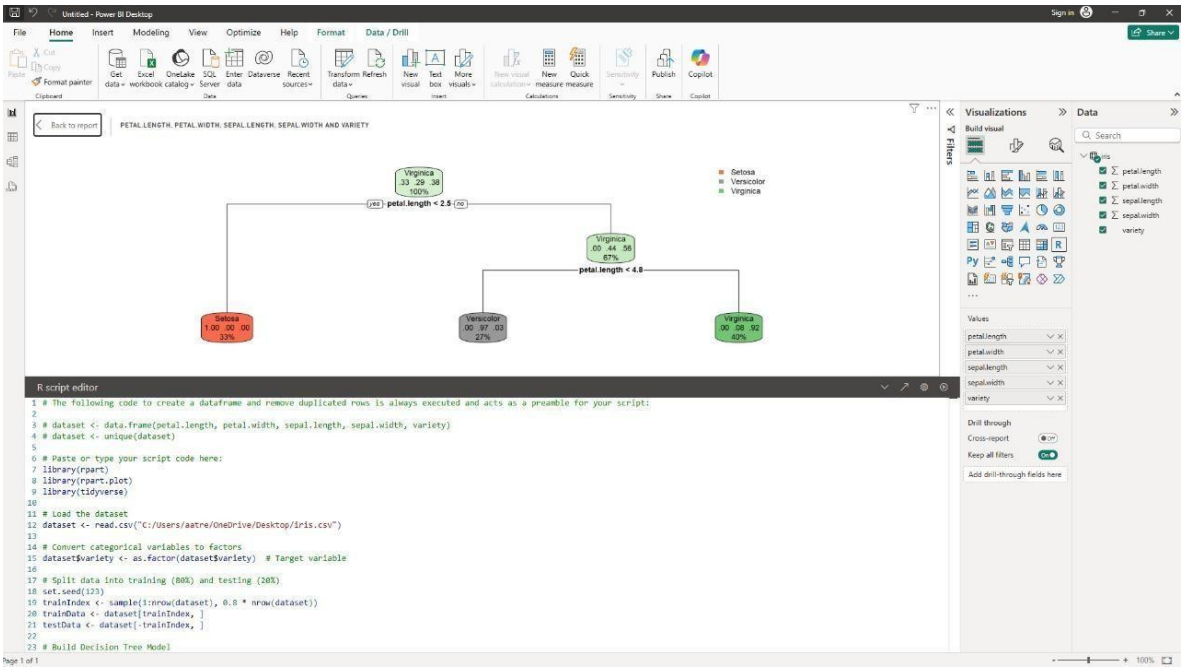
Step 3: Select No of Parameters from the dataset to work within it.



Step 4: In the Script editor Type the Decision Tree algorithm using R for visualization



Step 5: Run the script for the visualization of the code

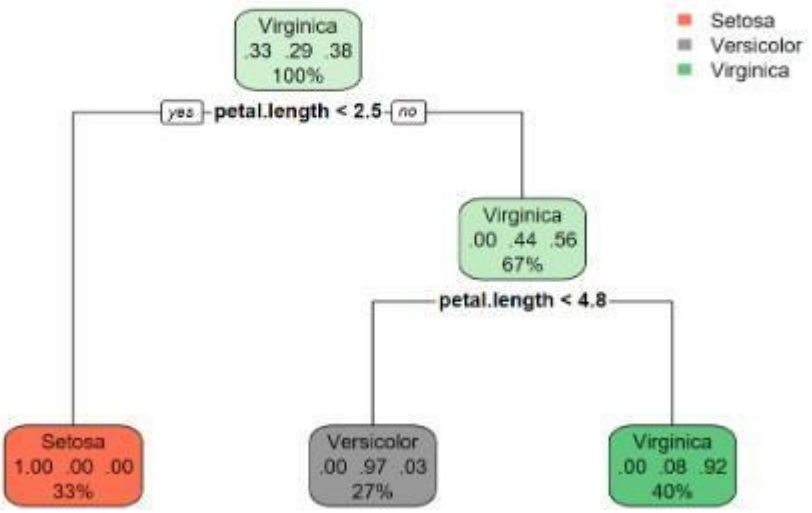


CODE:

```
library(rpart)
library(rpart.plot)
library(tidyverse)
# Load the dataset
dataset <- read.csv("C:/Users/AI_LAB/Desktop/iris.csv")
# Convert categorical variables to factors
dataset$variety <- as.factor(dataset$variety)  # Target variable
# Split data into training (80%) and testing (20%)
set.seed(123)
trainIndex <- sample(1:nrow(dataset), 0.8 * nrow(dataset))
trainData <- dataset[trainIndex, ]
testData <- dataset[-trainIndex, ]
# Build Decision Tree Model
dt_model <- rpart(variety ~ ., data = trainData, method = "class")
# Plot the Decision Tree
rpart.plot(dt_model)
predictions <- predict(dt_model, testData, type = "class")
# Evaluate model performance
confMatrix <- confusionMatrix(predictions, testData$Species)
print(confMatrix)
```

OUTPUT:

Sum of petal.length, Sum of petal.width, sepal.width, sepal.length and variety



RESULT:

Thus, the implementation of Decision Tree using R programming in Power BI was executed successfully and the output also verified.